

- Q.27 Explain the different human factors to be considered while designing a product.
- Q.28 What is meant by Industrial disputes? Explain its causes and effects?
- Q.29 Define Incentives and its types?
- Q.30 Define product development? Explain its different steps.
- Q.31 Define Training? Explain its objectives.
- Q.32 Explain the standard data.
- Q.33 How productivity is measured? Explain.
- Q.34 Explain why do we need inventory of items?
- Q.35 Describe the role of supervisor in Industrial Engineering?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Make a Diagram of PPC by suggesting its various examples with advantages and functions.
- Q.37 Define Route sheets? Explain various stages of PPC with suitable examples?
- Q.38 Explain with line diagram of Human aspects of work study and also define its role in improving productivity.

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4th Semester/ Mech Subject: Industrial Engineering

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Productivity is the _____ of production system
a) Measurement b) Efficiency
c) Both A & B d) None
- Q.2 Work-study related to
a) Changing the current procedure
b) To modernize
c) Increasing productivity
d) Expanding the industry
- Q.3 After selection of procedure according to established procedure for study of method, next step is
a) Recording b) Test
c) Development d) Establishment
- Q.4 For recording movements of work in micro motion study
a) Motion picture camera b) Microscope
c) Projector d) Stopwatch

- Q.5 What describes therblig in micromotion study?
- A symbol
 - A colour
 - An event
 - Standard symbol and colour
- Q.6 Elimination of which of following will improve industrial productivity
- Excess time
 - Idle time
 - Ineffective time
 - Work content
- Q.7 What is Productivity?
- Working on products
 - Goods that are produced
 - Amounts of goods or services
 - Workers can produce in a period of time
- Q.8 Which is not used in Method Study?
- Flow Diagram
 - Cycle graph
 - Travel diagram
 - Stop watch study
- Q.9 Time scale used in which chart?
- Outline process
 - Flow process
 - Man-Machine
 - Travel
- Q.10 Gantt charts are associated with
- Material handling
 - Inventory control
 - Sales fore-cast
 - Production schedule

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Write an example of productivity?
- Q.12 What is work study job?
- Q.13 What is Information collection in Method study?
- Q.14 Define stopwatch time study?
- Q.15 Define Allowance?
- Q.16 What is work sampling?
- Q.17 What is Incentive?
- Q.18 Define wage payment?
- Q.19 Give an example of production?
- Q.20 Write full form of EOQ?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain Contribution of standardization in improving productivity?
- Q.22 Explain the factors for selection of work-study job?
- Q.23 Explain the principle of Motion Economy?
- Q.24 Explain various Allowances in work Measurement?
- Q.25 Explain Job order type production with suitable example.
- Q.26 Explain any five fields of applications of Industrial Engineering?